



Monitoring the progression and pre-clinical diagnosis of Parkinson's disease by detection of α -synuclein oligomers in CSF using the PMCA assay

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Summary

The α -synuclein Protein Misfolding Cyclic Amplification (α S-PMCA) assay takes advantage of the intrinsic self-replicating nature of α -Synuclein (α -Syn) misfolded aggregates, and cyclically multiplies this biomarker for Parkinson's disease (PD). Amplification occurs *in vitro* by fragmenting the endogenous CSF α Syn aggregates by shaking the reaction at high speed. The fragmentation generates more and smaller seeds, which will then grow by incorporation of native recombinant protein into the aggregate. Given the amplificative nature of the technique, minute amounts of misfolded protein aggregates circulating in human bodily fluids can be multiplied and thereafter detected by conventional methods, such as immune-assays or fluorescence. Following these principles, α S-PMCA was standardized for the highly sensitive and specific detection of α Syn misfolded aggregates in cerebrospinal fluid (CSF) of patients with PD (1–3).

Method

Slow- α S-PMCA

The slow α S-PMCA reaction for amplification of endogenous α Syn aggregates in CSF is 100 mM PIPES pH 6.5, 0.5 M NaCl, 10 μ M ThT, 1 mg/mL recombinant α Syn, and 20% CSF v/v. The reaction is performed in ELISA plates and the samples are analyzed in triplicates. The plate is incubated at 37°C and shaken for 1 min every 29 min. Fluorescence is read at 490nm-Em/440nm-Ex after every shaking cycle for 600 cycles (300h, 12.5 days). CSF samples containing α Syn aggregates associated to PD display aggregation curves within 300h, while CSF samples from control patients do not show aggregation. In this project, 139 unidentified CSF samples were analyzed using the slow α S-PMCA assay.

Fast- α S-PMCA

The fast α S-PMCA reaction for amplification of endogenous α Syn aggregates in CSF is 100 mM PIPES pH 6.5, 0.5 M NaCl, 10 μ M ThT, 0.3 mg/mL recombinant α Syn, 20% CSF v/v and 1 silicon nitride bead. The reaction is performed in ELISA plates and the samples are analyzed in triplicates. The plate is incubated at 37°C and shaken for 1 min every 29 min. Fluorescence is read at 490nm-Em/440nm-Ex after every shaking cycle for 300 cycles (150h, 6.2 days). CSF samples containing α Syn aggregates associated to PD display aggregation curves within 150h,





while CSF samples from control patients do not show aggregation. In this project, 140 unidentified CSF samples were analyzed using the fast α S-PMCA assay. 139 of these samples were also analyzed with the slow assay and the results compared.

Data analysis

The 490nm fluorescence values are graphed vs time. α S-PMCA positive CSF samples normally present a classic amyloid aggregation curve with a lag phase, an exponential growth and a final plateau. α S-PMCA negative CSF samples present background fluorescence values, but no aggregation curve. Parameters like maximum fluorescence (Fmax), time to reach 50% of the maximum fluorescence (T50), and time to reach a threshold (TTT) can be calculated for the positive samples to characterize the aggregation curves. Except Fmax, these parameters cannot be calculated for negative samples. Therefore, the parameter Fmax is used to differentiate between negative and positive. Slow and fast assay have slightly different decision criteria.

Slow assay

Since CSF is a complex matrix and many factors can induce/inhibit aggregation of α Syn, the results of each replicate within a triplicate are combined to determine if the CSF sample was collected from a control or PD patient. The decision criteria for the slow assay consists of 2 steps, the first at the replicate level and the second at the triplicate level:

Decision criteria – replicate level

- If a replicate reaches Fmax above 1,000 RFU, the replicate is considered positive.
- If the Fmax of a replicate is less than 1,000 RFU, the replicate is considered negative.

Decision criteria – triplicate level

- PMCA positive (PD):
 - o If all three replicates aggregate (Fmax>1,000), the sample is considered PMCA positive (PD).
- PMCA Negative:
 - o If none or one of the three replicates show aggregation (Fmax>1,000), the CSF is considered negative and the patient is called control or non-PD.
- PMCA Inconclusive:
 - o If two of the replicates aggregate (Fmax>5,000), the sample is called inconclusive and there is no call respect to the status of the patient. Re-test is advised.

In this project, 139 CSF samples were analyzed with the slow assay. 39 were found positive, 83 negative, and 17 inconclusive. Some vials did not have enough volume to run the full CSF volume needed for the third replicate. It was decided to add whatever volume was available to the third replicate and report a qualitative measurement of the volume used in the third replicate. Samples labelled as “Not Enough Volume 1” (NEV1) are CSF samples that contained between 50-75% of the volume required for the third replicate. Samples labelled as “Not Enough Volume





2" (NEV2) are samples that contained more than 75% of the sample required for the third replicate but were still short in volume.

Fast assay

The classic decision criteria for the fast assay consists of 2 steps, the first at the replicate level and the second one at the triplicate level:

Decision criteria – replicate level

- If a replicate reaches Fmax above 5,000 RFU, the replicate is considered positive.
- If the Fmax of a replicate is less than 5,000 RFU, the replicate is considered negative.

Decision criteria – triplicate level

- PMCA positive (PD):
 - o If all three replicates aggregate (Fmax>5,000), the sample is considered PMCA positive (PD).
- PMCA Negative:
 - o If none or one of the three replicates show aggregation (Fmax>5,000), the CSF is considered negative and the patient is called control or non-PD.
- PMCA Inconclusive:
 - o If two of the replicates aggregate (Fmax>5,000), the sample is called inconclusive and there is no call respect to the status of the patient. Re-test is advised.

In this project, 140 samples were analyzed with the fast assay. 59 were positive, 72 were negative and 9 were inconclusive. Some vials did not have enough volume to run the full CSF volume needed for the third replicate. It was decided to add whatever volume was available to the third replicate and report a qualitative measurement of the volume used in the third replicate. Samples labelled as "Not Enough Volume 1" (NEV1) are CSF samples that contained between 50-75% of the volume required for the third replicate. Samples labelled as "Not Enough Volume 2" (NEV2) are samples that contained more than 75% of the sample required for the third replicate but were still short in volume.

Fast assay – Algorithm

Amprion generated a database containing aggregation parameters for approximately 1,000 tests of PD and Control samples analyzed with the fast assay. Using this database, a proprietary algorithm was developed to increase accuracy of the test by introducing more kinetic parameters to the decision criteria. With these parameters, the algorithm determines the probability of each replicate to belong to a PD or control sample. As second step, the results of all 3 replicates are combined to determine if the samples are positive, negative or inconclusive.

The results of the samples analyzed with the fast assay were also analyzed with the new algorithm as decision criteria. Of the 140 samples analyzed with the fast assay and the algorithm, 59 were positive, 78 negative and 3 inconclusive.





References

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